

Parallel Decision Making & Conflict Resolution



State/Event



Agent



Action/Q-value



Conflict



Resolution



Outcome

Event triggers decision at t_d

Freeze global state s_{t_d}

Stage 1: Parallel Action Selection

Agent J_3

Q-network
 $Q=4.2$

$a=M_2$

Agent J_7

Q-network
 $Q=3.8$

$a=M_2$

Agent J_{11}

Q-network
 $Q=2.5$

$a=M_1$

Stage 2: Conflict Detection

Collect: $\{J_3 \rightarrow M_2, J_7 \rightarrow M_2, J_{11} \rightarrow M_1\}$

Detect: $C_{M_2} = \{J_3, J_7\}$
 $|C_{M_2}| = 2 \rightarrow \text{CONFLICT!}$

M_2 conflict

M_1 no conflict

Stage 3: ϵ -greedy Resolution

Exploit: $\text{argmax } Q \rightarrow J_3 \text{ wins}$
($4.2 > 3.8$)

Winner
 J_3 gets M_2

Loser

J_7 waits till next
decision point
waiting time penalty

TD Update:
 $Q(o^7, M_2) \downarrow$

J_{11} gets M_1
(success)